

10.3 Unit Guide

In this unit, the handbook treats you like an idiot, so there will be minimal calculations on your end. As a result, no calculator exists, and there is only really one type of question for me to guide you through, so it will be rather quick.

On the other hand, I will try my best to make this walk-through as detailed as possible to help you understand every bit.

10.3.1 General Question Guide

Straight off the bat, the question itself should provide you with some basic information, such as:

- Operating pressure⁵⁷ [psi]
- Stroke Length [in.]
- Friction Coefficient
- Piston/Rod Diameters [in.]
- Acceleration/Deceleration condition⁵⁸
- Extension Speed [ft/min or in/sec]
- Pump Flow Rate [gpm]
- Maximum Load⁵⁹ [lbf]

You will likely not get all of these, but most design questions will definitely include the working pressure and stroke length to start you off.

Now, in most conventional fluid power design questions, you will need to determine the following:

- Sizing the Piston/Rod (if not given)
- Basic Length
- Flow Rate (if not given)
- Pipe diameter
- Motor Type/Size

Below are steps to solving a fluid power question from scratch.

⁵⁷This pressure is what the relief valve is set to

⁵⁸For example: "Your load must be accelerated to X ft/sec in X inches..."

⁵⁹Either given or found using statics

1. First, start off by writing down what you have been given, which will likely be all or some of the following:
 - Relief Pressure⁶⁰
 - Stroke Length
 - Acceleration/Deceleration condition
 - Extension Speed [ft/min or in/sec]
2. Next, depending on whether this is given or not, determine the maximum load on your cylinder. You might need to do some trig if the load is being pushed up a ramp at an angle or something like that.

Once you do that, use the table below to choose a piston (bore) with a stroke force that exceeds your maximum force rating. Remember to use your operating pressure!⁶¹

CUSTOMARY U.S. UNITS												
Cyl. Bore or Piston Rod Dia. (in.)	Cyl. Bore Size (φ mm)	Area (sq. in.)	CYLINDER PUSH STROKE FORCE IN POUNDS AT VARIOUS PRESSURES (PSI)									
			50	80	100	500	750	1000	1500	2000	2500	3000
5/8	15.9	.307	15	25	31	154	230	307	461	614	768	921
1	25.4	.785	39	65	79	392	588	785	1,177	1,570	1,962	2,355
1-3/8	34.9	1.490	75	119	149	745	1,118	1,490	2,235	2,980	3,725	4,470
1-1/2	38.1	1.767	88	142	177	885	1,325	1,770	2,651	3,540	4,425	5,310
1-3/4	44.5	2.410	121	193	241	1,205	1,808	2,410	3,615	4,820	6,025	7,230
2	50.8	3.140	157	251	314	1,570	2,357	3,140	4,713	6,280	7,850	9,420
2-1/2	63.5	4.910	245	393	491	2,455	3,682	4,910	7,364	9,820	12,275	14,730
3	76.2	7.070	354	566	707	3,535	5,302	7,070	10,604	14,140	17,675	21,210
3-1/4	82.6	8.300	415	664	830	4,150	6,225	8,300	12,450	16,600	20,750	24,900
3-1/2	88.9	9.620	481	770	962	4,810	7,215	9,620	14,430	19,240	24,050	28,860
4	101.6	12.570	628	1,006	1,257	6,285	9,428	12,570	18,856	25,140	31,425	37,710
5	127.0	19.640	982	1,571	1,964	9,820	14,730	19,640	29,460	39,280	49,100	58,920
5-1/2	139.7	23.760	1,188	1,901	2,376	11,880	17,820	23,760	35,640	47,520	59,400	71,280
6	152.4	28.270	1,414	2,262	2,827	14,135	21,203	28,270	42,406	56,540	70,675	84,810
7	177.8	38.490	1,924	3,079	3,849	19,245	28,868	38,490	57,736	76,980	96,225	115,470
8	203.2	50.270	2,513	4,022	5,027	25,135	37,703	50,270	75,406	100,540	125,675	150,810
8-1/2	215.9	56.750	2,838	4,540	5,675	28,375	42,563	56,750	85,125	113,500	142,875	170,250
10	254.0	78.540	3,927	6,283	7,854	39,270	58,905	78,540	117,810	157,080	196,350	235,620
12	304.8	113.100	5,655	9,048	11,310	56,550	84,825	113,100	169,650	226,200	282,750	339,300

table b-1

NOTE: Deduct Force of Piston Rod Size from Bore Size for Pull Applications.

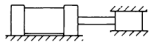
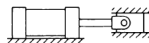
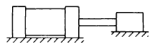
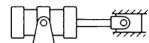
3. Find the basic (effective) length, given by:

$$\text{Basic Length} = \text{Actual Stroke} \times \text{Stroke Factor}$$

The actual stroke is given through the question, and the stroke factor is found based on the mounting style and rod end connection using the table below.

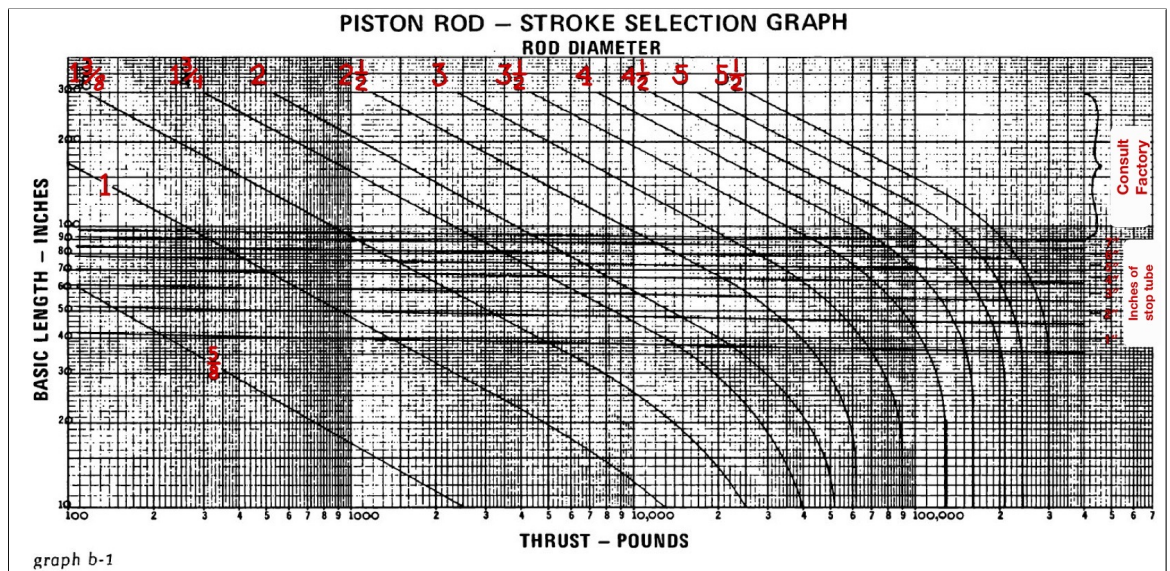
⁶⁰If not given, I'm pretty sure you just assume 1000 psi

⁶¹There's a mistake in one of the force values in this graph, let's see if one of you can find it

piston rod — stroke selection table			
RECOMMENDED MOUNTING STYLES FOR MAXIMUM STROKE AND THRUST LOADS	ROD END CONNECTION	CASE	STROKE FACTOR
CLASS 1 — GROUPS 1 OR 3 Long stroke cylinders for thrust loads should be mounted using a heavy-duty mounting style at one end, firmly fixed and aligned to take the principle force. Additional mounting should be specified at the opposite end, which should be used for alignment and support. An intermediate support may also be desirable for long stroke cylinders mounted horizontally. Machine mounting pads can be adjustable for support mountings to achieve proper alignment.	FIXED AND RIGIDLY GUIDED.	I 	.50
	PIVOTED AND RIGIDLY GUIDED	II 	.70
	SUPPORTED BUT NOT RIGIDLY GUIDED	III 	2.00
CLASS 2 — GROUP 2 Style — Trunnion on Head	PIVOTED AND RIGIDLY GUIDED	IV 	1.00
Style — Intermediate Trunnion	PIVOTED AND RIGIDLY GUIDED	V 	1.50
Style — Trunnion on Cap or Style — Clevis on Cap	PIVOTED AND RIGIDLY GUIDED	VI 	2.00

4. Choose a rod diameter using the graph⁶² below (round up!).⁶³

"Thrust" is the maximum force that is given or calculated and "Basic Length" is what we calculated in the previous step.



5. Once you choose both the cylinder bore and rod diameters, use the tables below to check that the following combo is available.

⁶²I TRIED to make it clearer. Don't worry about the stop tube stuff....

⁶³In-case the writings are too small, access the full size graphs [here](#)

CYLINDER BORE-INCHES	PISTON ROD		CYLINDER NET AREA SQ. IN.	FLUID DISPLACEMENT AT 10 FT. PER MINUTE PISTON VELOCITY		FLUID VELOCITY (IN FEET PER SECOND) THROUGH EXTRA HEAVY PIPE AT 10 F.P.M. PISTON SPEED.									
	DIA.-INCHES	AREA SQ. IN.													
						G.P.M.	C.F.M.	1/4	3/8	1/2	3/4	1	1-1/4	1-1/2	2
1	0	0	0.785	0.41	0.054	1.82	0.92	0.56	0.30	0.183	0.102	0.074	0.045		
	1/2	0.196	0.589	0.30	0.041	1.33	0.68	0.41	0.21	0.134	0.075	0.055	0.033		
	5/8	0.307	0.478	0.16	0.033	0.71	0.36	0.22	0.12	0.071	0.040	0.029	0.017		
1½	0	0	1.77	0.92	0.123	4.09	2.09	1.259	0.680	0.410	0.230	0.167	0.100		
	5/8	0.307	1.46	0.76	0.101	3.38	1.73	1.040	0.562	0.338	0.190	0.138	0.082		
	1	0.785	0.98	0.51	0.068	2.27	1.16	0.699	0.378	0.228	0.128	0.093	0.055		
2	0	0	3.14	1.63	0.218	7.27	3.71	2.238	1.209	0.728	0.408	0.296	0.177		
	5/8	0.307	2.84	1.48	0.197	6.56	3.35	2.019	1.091	0.657	0.368	0.267	0.160		
	1	0.785	2.36	1.23	0.164	5.45	2.79	1.678	0.907	0.546	0.306	0.222	0.133		
2½	1-3/8	1.485	1.66	0.86	0.115	3.84	1.96	1.180	0.638	0.384	0.215	0.156	0.094		
	0	0	4.91	2.55	0.341	11.36	5.80	3.496	1.890	1.138	0.638	0.463	0.277		
	5/8	0.307	4.60	2.39	0.319	10.65	5.44	3.278	1.771	1.067	0.598	0.434	0.260		
3	1	0.785	4.12	2.14	0.286	9.54	4.87	2.937	1.587	0.956	0.536	0.389	0.233		
	1-3/8	1.485	3.42	1.78	0.237	7.93	4.05	2.439	1.318	0.794	0.445	0.323	0.193		
	1-3/4	2.405	2.50	1.30	0.174	5.96	2.96	1.783	0.963	0.580	0.325	0.236	0.141		
3¾	0	0	8.30	4.31	0.576	19.20	9.81	5.909	3.193	1.923	1.078	0.783	0.468		
	1	0.785	7.51	3.90	0.521	17.38	8.88	5.349	2.891	1.741	0.976	0.708	0.424		
	1-3/8	1.485	6.81	3.54	0.473	15.77	8.05	4.851	2.622	1.579	0.885	0.642	0.384		
4	1-3/4	2.405	5.89	3.06	0.409	13.64	6.96	4.196	2.268	1.366	0.765	0.556	0.333		
	2	3.142	5.15	2.68	0.357	11.93	6.09	3.671	1.984	1.195	0.670	0.486	0.291		
	0	0	12.57	6.53	0.872	29.09	14.85	8.95	4.84	2.91	1.63	1.19	0.709		
4½	1	0.785	11.78	6.12	0.818	27.27	13.93	8.39	4.54	2.73	1.53	1.11	0.665		
	1-3/8	1.485	11.08	5.76	0.769	25.65	13.10	7.89	4.27	2.57	1.44	1.05	0.625		
	1-3/4	2.405	10.16	5.28	0.705	23.52	12.01	7.24	3.91	2.36	1.32	0.96	0.574		
5	2	3.142	9.42	4.89	0.654	21.82	11.14	6.71	3.63	2.19	1.22	0.89	0.532		
	2-1/2	4.909	7.66	3.98	0.532	17.73	9.05	5.45	2.95	1.78	1.00	0.72	0.432		
	0	0	19.64	10.20	1.363	45.45	23.21	13.99	7.56	4.55	2.55	1.85	1.108		
5½	1	0.785	18.85	9.79	1.308	43.64	22.28	13.43	7.26	4.37	2.45	1.78	1.064		
	1-3/8	1.485	18.15	9.43	1.260	42.01	21.45	12.93	6.99	4.21	2.36	1.71	1.024		
	1-3/4	2.405	17.23	8.95	1.196	39.88	20.37	12.27	6.63	3.99	2.24	1.63	0.973		
6	2	3.142	16.49	8.57	1.144	38.18	19.50	11.75	6.35	3.82	2.14	1.56	0.931		
	2-1/2	4.909	14.73	7.65	1.022	34.09	17.41	10.49	5.67	3.41	1.91	1.39	0.831		
	3	7.069	12.57	6.53	0.872	29.09	14.85	8.95	4.84	2.91	1.63	1.19	0.709		
6½	3-1/2	9.621	10.01	5.21	0.695	23.18	11.84	7.13	3.86	2.32	1.30	0.95	0.565		
	0	0	28.27	14.69	1.962	65.45	33.42	20.14	10.88	6.55	3.67	2.67	1.596		
	1-3/8	1.485	26.79	13.92	1.859	62.01	31.67	19.08	10.31	6.21	3.48	2.53	1.512		
7	1-3/4	2.405	25.87	13.44	1.795	59.88	30.58	18.43	9.96	5.60	3.36	2.44	1.460		
	2	3.142	25.13	13.06	1.744	58.18	29.71	17.90	9.67	5.83	3.27	2.37	1.418		
	2-1/2	4.909	23.37	12.14	1.622	54.1	27.6	16.64	8.99	5.42	3.04	2.20	1.32		
7½	3	7.069	21.21	11.02	1.472	49.1	25.1	15.10	8.16	4.92	2.76	2.00	1.20		
	3-1/2	9.621	18.65	9.69	1.294	43.2	22.1	13.29	7.18	4.32	2.42	1.76	1.05		
	4	12.566	15.71	8.16	1.090	36.4	18.6	11.19	6.05	3.64	2.04	1.48	0.89		

table b-5

CYLINDER BORE-INCHES	PISTON ROD		CYLINDER NET AREA SQ. IN.	FLUID DISPLACEMENT AT 10 FT. PER MINUTE PISTON VELOCITY		FLUID VELOCITY (IN FEET PER SECOND) THROUGH EXTRA HEAVY PIPE AT 10 F.P.M. PISTON SPEED.									
	DIA.-INCHES	AREA SQ. IN.		G.P.M.	C.F.M.	1/4	3/8	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	
7	0	0	38.49	20.00	2.671	89.1	45.5	27.41	14.81	8.92	5.00	3.63	2.17		
	1-3/8	1.485	37.00	19.22	2.568	85.7	43.7	26.35	14.24	8.58	4.81	3.49	2.09		
	1-3/4	2.405	36.08	18.74	2.504	83.5	42.7	25.70	13.89	8.36	4.69	3.40	2.04		
	2	3.142	35.34	18.36	2.453	81.8	41.8	25.17	13.60	8.19	4.59	3.33	2.00		
	2-1/2	4.909	33.58	17.44	2.330	77.7	39.7	23.92	12.92	7.78	4.36	3.17	1.90		
	3	7.069	31.42	16.32	2.181	72.7	37.1	22.38	12.09	7.28	4.08	2.96	1.77		
	3-1/2	9.621	28.86	14.99	2.003	66.8	34.1	20.56	11.11	6.69	3.75	2.72	1.63		
	4	12.566	25.92	13.47	1.799	60.0	30.6	18.46	9.98	6.01	3.37	2.45	1.46		
	4-1/2	15.904	22.58	11.73	1.567	52.3	26.7	16.08	8.69	5.23	2.93	2.12	1.28		
	5	19.635	18.85	9.79	1.308	43.6	22.3	13.43	7.26	4.37	2.45	1.78	1.06		
8	0	0	50.27	26.12	3.489	116.4	59.4	35.80	19.35	11.65	6.53	4.74	2.84	1.977	
	1-3/8	1.485	48.78	25.34	3.385	112.9	57.7	34.74	18.78	11.31	6.34	4.60	2.75	1.918	
	1-3/4	2.405	47.86	24.86	3.321	110.8	56.6	34.09	18.42	11.09	6.22	4.51	2.70	1.882	
	2	3.142	47.12	24.48	3.270	109.1	55.7	33.56	18.14	10.92	6.12	4.45	2.66	1.853	
	2-1/2	4.909	45.36	23.57	3.149	105.0	53.61	32.31	17.46	10.51	5.892	4.278	2.560	1.784	
	3	7.069	43.20	22.44	2.998	100.0	51.06	30.77	16.63	10.01	5.612	4.074	2.438	1.699	
	3-1/2	9.621	40.65	21.12	2.821	94.1	48.04	28.95	15.65	9.42	5.279	3.834	2.294	1.598	
	4	12.566	37.70	19.59	2.616	87.3	44.56	26.85	14.51	8.74	4.897	3.556	2.128	1.483	
	4-1/2	15.904	34.36	17.85	2.385	79.5	40.62	24.47	13.23	8.20	4.464	3.241	1.939	1.351	
	5	19.635	30.63	15.91	2.126	70.9	36.21	21.82	11.79	7.10	3.979	2.889	1.729	1.205	
5-1/2	23.758	26.51	13.77	1.840	61.4	31.33	18.88	10.20	6.15	3.444	2.500	1.496	1.043		
10	0	0	78.54	40.80	5.451	181.8	92.84	55.94	30.23	18.21	10.203	7.408	4.433	3.089	
	1-3/4	2.405	76.14	39.56	5.284	176.2	89.99	54.23	29.31	17.65	9.890	7.181	4.297	2.994	
	2	3.142	75.40	39.17	5.233	174.5	89.12	53.70	29.02	17.48	9.795	7.112	4.255	2.965	
	2-1/2	4.909	73.63	38.25	5.110	170.4	87.03	52.44	28.34	17.07	9.565	6.945	4.156	2.896	
	3	7.069	71.47	37.13	4.960	165.4	84.48	50.91	27.51	16.57	9.284	6.741	4.034	2.811	
	3-1/2	9.621	68.92	35.80	4.783	159.5	81.47	49.09	26.53	15.98	8.953	6.501	3.890	2.710	
	4	12.566	65.97	34.27	4.578	152.7	77.98	46.99	25.39	15.29	8.570	6.223	3.724	2.595	
	4-1/2	15.904	62.64	32.54	4.347	145.0	74.04	44.61	24.11	14.52	8.137	5.908	3.535	2.463	
	5	19.635	58.91	30.60	4.088	136.4	69.63	41.96	22.67	13.65	7.652	5.556	3.325	2.317	
	5-1/2	23.758	54.78	28.46	3.802	126.8	64.75	39.02	21.09	12.70	7.116	5.167	3.092	2.154	
12	6	28.274	50.27	26.12	3.489	116.4	59.42	35.80	19.35	11.65	6.530	4.741	2.837	1.977	
	6-1/2	33.183	45.36	23.57	3.148	105.0	53.6	32.31	17.46	10.52	5.89	4.278	2.560	1.784	
	7	38.485	40.06	20.81	2.780	92.7	47.4	28.53	15.42	9.29	5.20	3.778	2.261	1.575	
	0	0	113.10	58.76	7.849	261.8	133.7	80.55	43.53	26.22	14.69	10.668	6.383	4.448	
	2	3.142	109.96	57.12	7.631	254.5	130.0	78.32	42.32	25.49	14.28	10.371	6.206	4.324	
	2-1/2	4.909	108.19	56.21	7.508	250.4	127.9	77.06	41.64	25.08	14.05	10.205	6.106	4.255	
	3	7.069	106.03	55.08	7.359	245.4	125.3	75.52	40.81	24.58	13.77	10.001	5.984	4.170	
	3-1/2	9.621	103.48	53.76	7.182	239.5	122.3	73.70	39.83	23.99	13.44	9.760	5.840	4.069	
	4	12.566	100.53	52.23	6.977	232.7	118.8	71.60	38.70	23.30	13.06	9.482	5.674	3.954	
	4-1/2	15.904	97.19	50.49	6.745	225.0	114.9	69.23	37.41	22.53	12.63	9.168	5.486	3.822	
14	5	19.635	93.46	48.55	6.486	216.4	110.5	66.57	35.98	21.67	12.14	8.816	5.275	3.676	
	5-1/2	23.758	89.34	46.41	6.200	206.8	105.6	63.63	34.39	20.71	11.61	8.427	5.042	3.513	
	6	28.274	84.82	44.06	5.887	196.4	100.3	60.42	32.65	19.66	11.02	8.001	4.787	3.336	
	6-1/2	33.183	79.92	41.52	5.547	185.0	94.5	56.92	30.76	18.53	10.38	7.538	4.510	3.143	
	7	38.485	74.61	38.77	5.179	172.7	88.2	53.14	28.72	17.30	9.69	7.038	4.211	2.934	
	7-1/2	44.179	68.92	35.80	4.783	159.5	81.5	49.09	26.53	15.98	8.95	6.501	3.890	2.710	
	8	50.266	62.83	32.64	4.360	145.4	74.3	44.75	24.19	14.57	8.16	5.926	3.546	2.471	
	8-1/2	56.745	56.35	29.27	3.911	130.5	66.6	40.14	21.69	13.06	7.32	5.315	3.181	2.216	
	0	0	153.94	79.97	10.683	356.3	182.0	109.6	59.25	35.68	20.00	14.52	8.688	6.054	
	2-1/2	4.909	149.03	77.42	10.343	345.0	176.2	106.2	57.36	34.55	19.36	14.06	8.411	5.861	

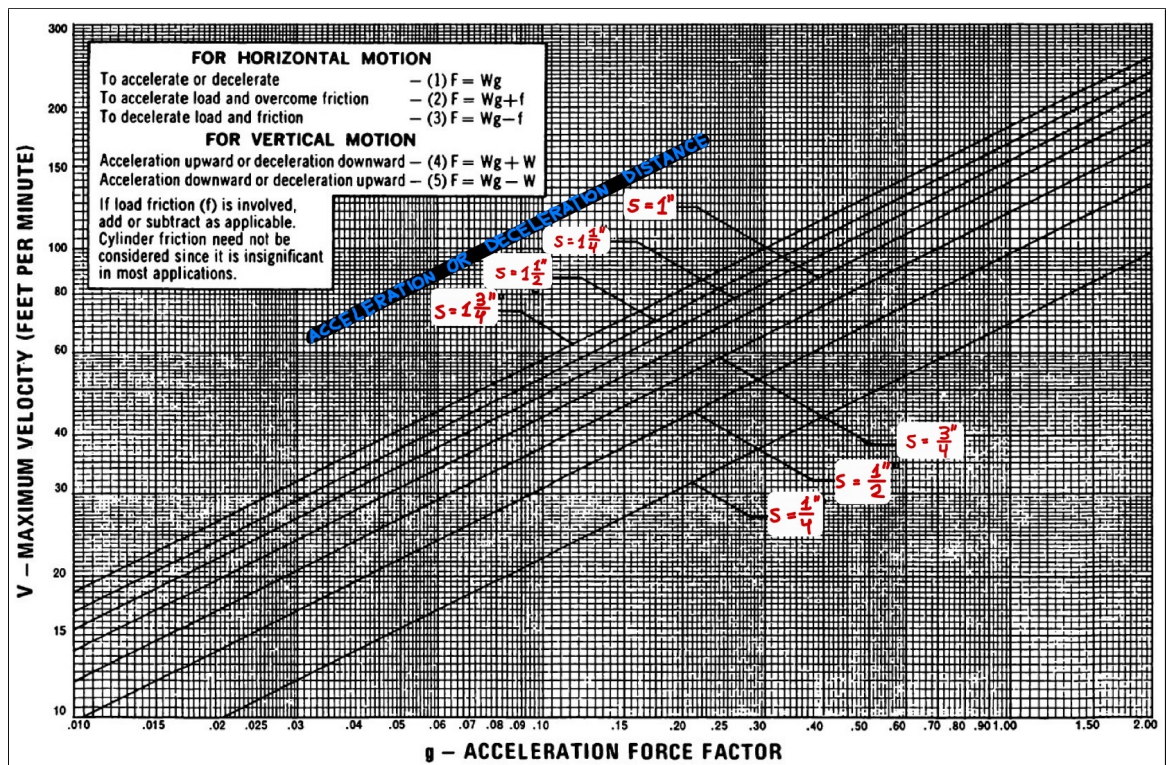
**If the rod is not available for the bore that you chose, choosing a larger bore and using the same rod works most of the time*

6. Check the acceleration/deceleration condition against your cylinder/rod selection. Use the graph below to find the value of the Acceleration Force Factor "g" and calculate the total force to make sure it doesn't exceed the allowed bore rating. The way to calculate F is given for some conditions in the top-left portion of the below graph.

If you prefer a more reliable result, use the equation below instead:

$$\frac{V^2}{S} \times 0.0000517$$

Here, V is the velocity you need to accelerate to, and S is the distance the acceleration must occur over.



7. Find the flow rate [GPM]. Use the given extension speed (convert to ft/min if needed) and find the flow rate [GPM] using the tables above in step 5 (use your piston diameter with 0 inch rod diameter for extension problems!).⁶⁴

The tables present the flow rate at 10 ft/min piston velocity, therefore, use this equation to get the "true value" for your piston speed [ft/min]:

$$Q_{true} = \frac{Piston\ Speed}{10} \times Q_{table}$$

This will be the flow rate required for your piping!

8. Choose Pipe Diameter. Again, use the tables in step 5, which give the fluid velocity for different pipe diameters at 10 ft/min piston speed (Use 0 rod diameter for extension case!). Make sure the fluid velocity is less than 15 ft/sec. In other words:

$$Fluid\ Velocity \times \frac{Piston\ Speed}{10} < 15 ft/s$$

In some cases you may be able to calculate it, but I suggest just sticking to the table and choosing the smallest piping possible (to cut down on costs of course!).

9. Next, you must choose a motor. First, find your minimum horsepower. Now there is a table to look at below, but I'd suggest just using the equation below:

$$HP_{min} = \frac{GPM \times PSI}{1714 \times 0.85}$$

Then, using the minimum horsepower you calculated, choose a motor from the tables below (usually 3-phase).

⁶⁴Some questions may just give you the flow rate, so skip this step if that's the case

3 PHASE MOTOR STARTERS																									
1/2 TO 20 H.P.	MOTOR H.P. 3 ∅		1/2		3/4		1		1-1/2		2		3		5		7-1/2		10		15		20		
VOLTAGE		220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440
Nema Starter Size		00	00	00	00	00	00	00	00	0	00	0	0	1	0	1	1	2	1	2	2	3	2		
⊕ Full Load Current		2.0	1.0	2.8	1.4	3.5	1.8	5.0	2.5	6.5	3.3	9.0	4.5	15	7.5	22	11	28	14	40	20	52	26		
Fuses — Amps { Std. N.E.C.		15	15	15	15	15	15	15	15	20	15	25	15	40	20	60	30	70	35	100	50	150	70		
Dual Element		5	4	5	4	8	4	8	5	12	8	15	10	25	15	30	20	45	25	60	30	80	40		
Circuit Breaker Max. Amps.		15	15	15	15	15	15	15	15	15	15	20	15	30	15	50	20	50	30	70	40	100	50		
Minimum Wire Sizes { R, RW, T, TW		14	14	14	14	14	14	14	14	14	14	14	14	14	14	12	14	10	14	8	12	6	10	4	8
RH		14	14	14	14	14	14	14	14	14	14	14	14	12	14	10	14	8-	12	6	10	6	8		8
Always specify voltage and frequency.																									

3 PHASE MOTOR STARTERS																							
25 TO 200 H.P.	MOTOR H.P. 3 ∅		25		30		40		50		60		75		100		125		150		200		
VOLTAGE		220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440	220	440
Nema Starter Size		3	2	3	3	4	3	4	3	5	4	5	4	5	4	6	5	6	5	6	5	6	5
⊕ Full Load Current		64	32	78	39	104	52	125	63	150	75	185	93	246	123	310	155	360	180	480	240		
Fuses — Amps. Std. N.E.C.		175	80	200	100	300	150	350	175	400	200	500	250	600	350	—	400	—	450	—	600		
Dual Element		100	50	125	60	175	80	200	100	225	125	300	150	400	200	450	250	600	300	—	400		
Circuit Breaker Max. Amps.		125	-50	100	70	175	100	200	125	225	125	300	150	400	200	—	250	—	300	—	400		
Minimum Wire Sizes { R, RW, T, TW		3	8	1	6	00	4	000	3	0000	2	300	0	500	000	—	0000	—	300	—	500		
RH		4	8	3	6	1	6	00	4	000	3	0000	1	350	00	—	000	—	0000	—	350		
Always specify voltage and frequency.																							

SINGLE PHASE MOTOR STARTERS																							
1/6 TO 5 H.P.	MOTOR H.P. 1 ∅		1/6		1/4		1/3		1/2		3/4		1		1-1/2		2		3		5		
VOLTAGE		115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230	115	230
⊕ Full Load Current		4.4	2.2	5.8	2.9	7.2	3.6	9.8	4.9	13.8	6.9	16	8	20	10	24	12	34	17	56	28		
Fuses — Amps. Std. N.E.C.		15	15	20	15	25	15	30	15	45	25	50	25	60	30	80	40	100	60	—	90		
Circuit Breaker Max. Amps.		15	15	15	15	15	15	30	15	40	20	40	20	50	30	70	30	100	50	—	70		
Min. Wire Sizes — R, RH, RW, T, TW		14	14	14	14	14	14	14	14	12	14	12	14	10	14	10	14	6	10	—	8		

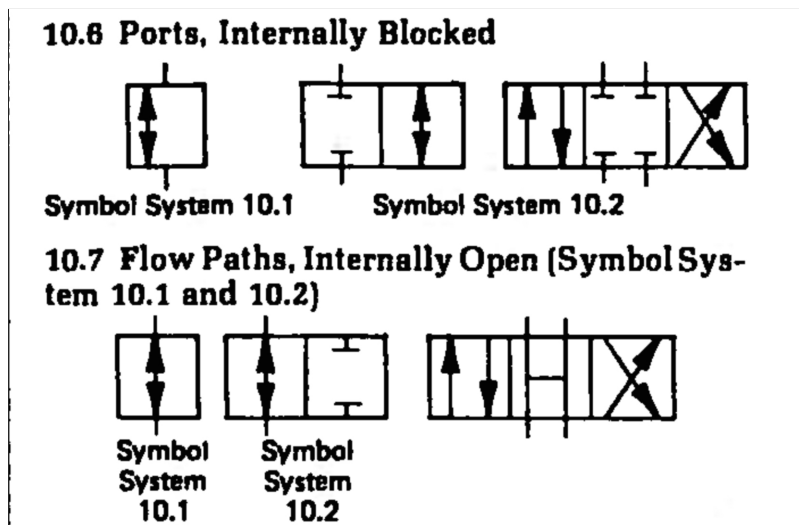
Don't worry too much about the voltages and other specs other than the HP, so choose them rather arbitrarily.

You should now have a fully specified system!
Good work!

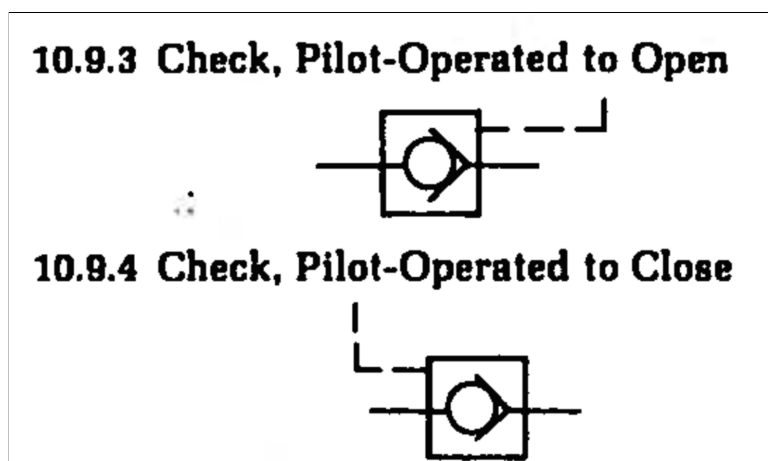
10.4 Figures, Definitions, and Relevant Questions

Below is an overview of fluid symbols, introducing what you'll generally need to know to understand what's happening when looking at a fluid circuit.⁶⁵

We will start off with valves. Valves are composed of one or more envelopes (square boxes) with lines inside that indicate fluid flow. **Directional Control Valves** are shown in the figure below, which you'll probably see in almost every fluid circuit you'll analyze.

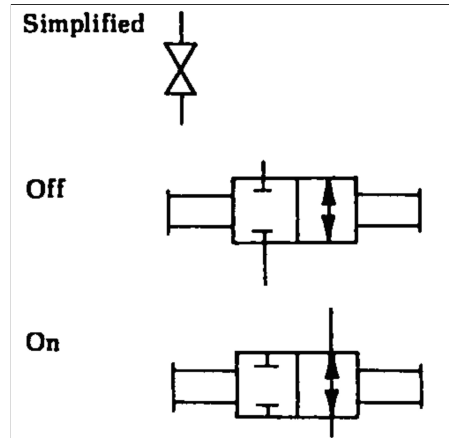


Check valves are valves where the flow to the right is blocked, but flow to the left is permitted. They are used to **prevent backflow**.

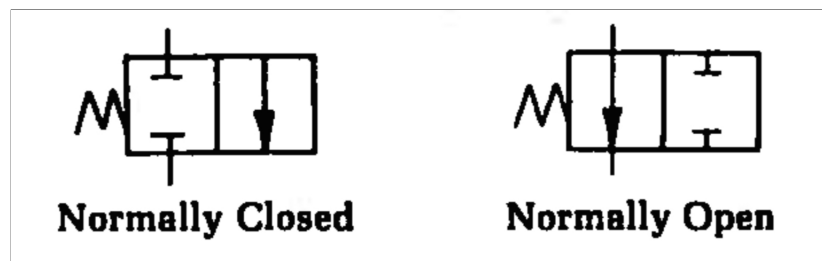


⁶⁵I will list good-to-know ones only

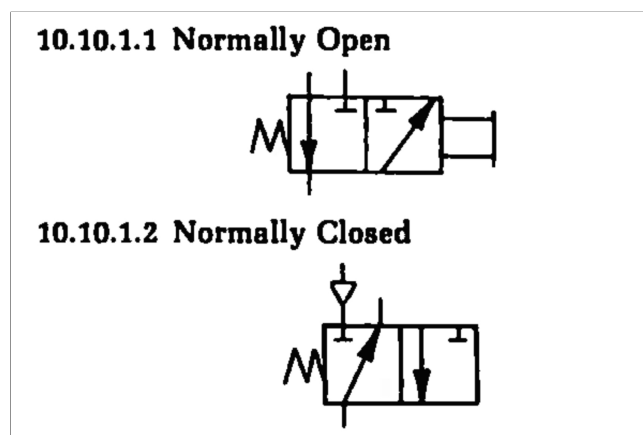
On-Off valves (manual) are as simple as it gets, either fluid flows or it doesn't.



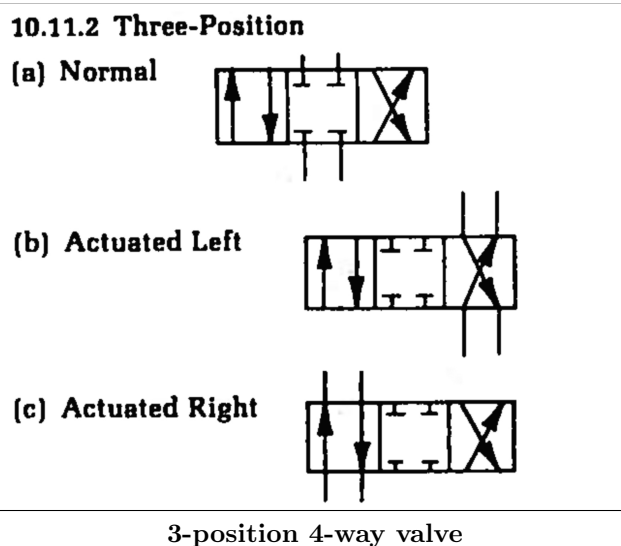
Two-way valves need not necessarily be manual. In some cases, a **spring** (squiggly line in the figure below) is used to switch between the two positions.



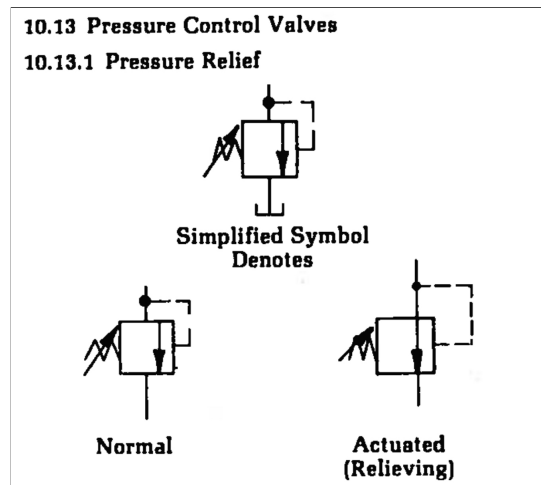
We can have 3-way or 4-way valves too. I have provided some examples below.



2-position 3-way valve

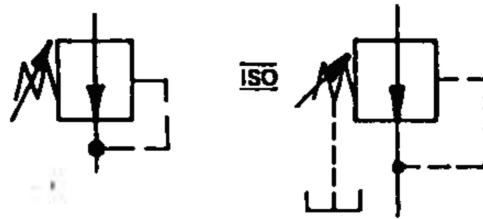


Now we move to **Pressure Control Valves**, and mainly the **Pressure Relief Valve**, which is used for safety and literally helps relief the system by diverting flow as the operating pressure is reached. This helps keep the system at a certain operating pressure.



Another pressure control valve is the **Pressure Reducing Valve**, which reduces the inlet pressure to a controlled and constant pressure at the outlet.

10.13.3 Pressure Reducing



Basically, the following valves should be enough for you to describe circuits in Mech 325 when asked to. There may be some that I missed, but I am sure the ones I've included are essential for your understanding of this unit.